

# Johan Ajnabi

PhD Candidate | Epithelial Biology, Mechanobiology and Epigenetics

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## WORK EXPERIENCE

### Institute for Stem Cell Science and Regenerative Medicine, India

Research Scholar ..... Aug 2019 – Present

#### ■Epigenetic and mechanical regulation of wound response:

Understanding the regulation of subcellular localization of *de novo* DNA methyltransferase, DNMT3a, an epigenetic regulator upon wound-induced mechanical cues (bioRxiv: [Ajnabi et al., 2026](#)).

#### ■Endocytosis-linked integrin signaling in keratinocyte stemness maintenance:

Uncovering a Mindin-integrin-STAT3 signaling axis that sustains keratinocyte stemness through integrin endocytosis and downstream transcriptional activation (Cell Communication and Signaling: [Dam et al., 2025](#)).

#### ■Spatial transcriptomic analysis of the immune landscape of vitiligo skin.

Profiled immune cell behavior in patient skin before and after narrow band UV-B treatment. (Clinical & Translational Immunology: [Dutta et al., 2026](#)).

#### ■Study of a matricellular protein in cutaneous fibrogenesis:

Understanding the role of Mindin as a critical regulator of dermal fibrogenesis in a mouse model of systemic sclerosis (Journal of Investigative Dermatology: [Rana et al., 2023](#)).

■Role of antimicrobial peptides (AMPs) in combating different variants of SARS-CoV2: Studying how niacinamide potentiates cathelicidin-mediated antiviral activity by enhancing membrane disruption of SARS-CoV-2 (Frontiers in Immunology: [Bhatt et al., 2023](#)).

■Maintenance of stem/progenitor state of skin epithelial cells and carcinomas: This study identifies an autocrine Snail-Mindin signaling loop that sustains epithelial stemness in both normal skin and carcinoma contexts (Cell Reports: [Badarinath et al., 2022](#)).

### ICAR-National Institute for Plant Biotechnology, India

AIEEA PG Scholar ..... Jul 2017 – Jul 2019

#### ■Identification of cis-regulatory region of a promoter:

Identifying cis-regulatory elements controlling PM19 gene expression in wheat, contributing to the understanding of stress-responsive transcriptional regulation. (Master's Thesis: [Ajnabi, 2019](#))

## FEATURED PUBLICATIONS (8 IN TOTAL)

■Ajnabi, J., Dam, B., Gupta, E., Saha, T., Dutta, A., Kumar, S., Gupta, A., Palakodeti, D., and Jamora, C., 2026. Actin-dependent mechanotransduction controls nucleocytoplasmic partitioning of DNMT3a through ERK1/2 signaling during cutaneous wound healing. bioRxiv, pp.2026-01.

## FEATURED PRESENTATIONS (5 IN TOTAL)

■Presented poster on “Epigenetic and mechanical regulation of the cutaneous wound healing” (MBI Conference 2023: Mechanobiology in Health and Disease, Singapore, September 26-29, 2023)

## FEATURED ACHIEVEMENTS

■Paeonia Travel Award for attending and presenting my work at the MBI Conference 2023: Mechanobiology in Health and Disease, NUS, Singapore

## SUMMARY

Scientist with expertise in mechanobiology, cell signaling, and epigenetic regulation. Experienced in mechanistic research using primary cell culture, mouse models, quantitative microscopy, flow cytometry, and molecular biology. Collaborative researcher with a strong publication record, seeking to advance translational research in regenerative medicine and cell biology.

## SKILLS

**Wet lab:** Handling and managing of mouse colonies and experimentation, primary cell culture, confocal and multiphoton microscopy, flow cytometry, genetic engineering, protein purification, BLSII practices, generation of pseudovirus, qPCR, western blots

**Software:** Basics of R, Python, MATLAB, MS Office, GraphPad Prism, Adobe Illustrator

**OS:** Linux, Windows

**Soft skills:** Responsible, organized, critical thinker, flexible, communicative, team player, patient

## EDUCATION

### BRIC-inStem ..... Bengaluru, India

PhD August 2019 – Present

Thesis: Regulation of DNMT3a localization in the cutaneous wound healing response.

Advisor: Dr. Colin Jamora

### ICAR-IARI ..... New Delhi, India

MSc in Molecular Biology and Biotechnology

July 2017 – July 2019

Thesis: Identification of cis-regulatory regions regulating the expression of PM19 gene in wheat.

Advisor: Dr. Monika Dalal GPA: 8.51/10

### BCKV ..... Mohanpur, India

BSc in Agriculture July 2013 – July 2017

GPA: 7.92/10 (with honours).

## LANGUAGES

■Bengali (Native) ■English (Advanced)

## INTERESTS

Coffee, Wildlife Photography, Piano, Hiking, Traveling